

EXAM 7 – SPRING 2013

18. (1.25 points)

A reinsurance company is pricing a casualty excess-of-loss reinsurance treaty that covers loss and allocated loss adjustment expense (ALAE).

a. (0.75 point)

Discuss the relationship between loss and ALAE amounts for a casualty line of business, specifically, how the relationship typically differs between large and small claims.

b. (0.5 point)

Describe how the relationship between loss and ALAE can be modeled in the pricing of a casualty excess-of-loss reinsurance treaty.

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Question 18 Sample Answer

Solution 1

- a) Usually, the correlation between loss and ALAE is not the same for all claims amounts. For large claims, ALAE are usually large too because the insurer will try to pay less, which means higher attorney fees, higher expenses to investigate the claims, etc. For small claims, the correlation is harder to establish because the insurer will investigate all claims that could be frauds, and usually not spend much time and money on the other small claims. So, for small claims, you could have a great range of ALAE, from almost zero to many thousands.
- b) One method to do so is to use a copula. A copula describes the correlation between two sets of values, loss and ALAE for this example. For example, if the copula selected is the Heavy Right Tail, it will mean that for high losses the two distributions are correlated heavily. There are a lot of copula types, and the actuary has to test the data to find which one best describes the correlation.

Solution 2

- a) Loss & ALAE amounts can be correlated together especially when looking at the magnitude of claims. Large claims will likely require more ALAE as insurers are likely to fight these claims in court to drive down the overall size of the payment or not to pay at all. Small claims will likely incur little to no ALAE because an insurer will be more likely to pay the claim, but could also have claims where a lot of ALAE is spent and no loss has to be paid. Because of this, there will be less correlation between the two payments for small claims.
- b) Based on the nature of the relationship between loss & ALAE, I would include a copula in my model that reflects the high correlation in the tails. (Remember, large losses will have large ALAE.) I would further suggest using a Gumbel copula or Heavy Right Tail copula because both will have high correlation in tail.

Examiner Comment

- a) Most candidates were able to identify the positive correlation between loss & ALAE. Not so many were able to discuss the change in the correlation through the two distributions. Some candidates mentioned that you could have high ALAE on a zero loss through a successful claim defense, but were unable to tie that to a more general statement about the nature of the correlation. Many other candidates misunderstood the question to be about reinsurance treaty terms, discussing treating ALAE as part of loss or pro-rata with loss, and these responses earned no credit.
- b) Most candidates correctly identified the need for a copula, but many of these candidates failed to explain the features of a copula that make it suited for the purpose in question.